

Infinitesimal Calculus 2, Moed A

2023, Reichman university.

Do not write at the back of the papers!!

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Instructions:

1. Duration: 3 hours.

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The following exam has 5 questions. You must answer exactly 4 questions. Each question is worth 25 credits.

You must explain in full details all of your answers in all parts of the exam, and state all of the conditions for every theorem and/or claim. **Unexplained answers will not receive any credit. You do not have a choice between different items of the same question. It is forbidden to rely on any material that was not learned in the course. You must write your solutions only in the designated areas. Do not write at the back of the papers!**

1. (25 points)

- (a) (15 points) Prove that e is an irrational number. You may rely on any of the previous theorems or claims that were learned in the course.

Solution:

See lectures.

- (b) (10 points) Compute the value of the following definite integral

$$\int_0^{\pi^2} \cos(\sqrt{x}) dx$$

Solution:

Substituting $t = \sqrt{x}$ then $dt = \frac{1}{2\sqrt{x}} dx$ and hence $dx = 2\sqrt{x}dt = 2tdt$. Thus

$$\int_0^{\pi^2} \cos(\sqrt{x}) dx = 2 \int_0^{\pi} t \cos(t) dt = 2 \left[[t \sin t]_0^{\pi} - \int_0^{\pi} \sin t dt \right] = -2 \int_0^{\pi} \sin t dt = 2 [\cos t]_0^{\pi} = -4$$

2. (25 points)

- (a) (17 points)

- i. (2 points) Let $x_0 \in \mathbb{R}$ and let f be a function that is differentiable ∞ times at x_0 . Define the term: the Taylor series of f about x_0 .

Solution: The Taylor series of f about x_0 is defined as

$$T_f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} \cdot (x - x_0)^n$$

- ii. (10 points) Find the Taylor series about $x_0 = 0$ of the function $f(x) = \begin{cases} \frac{\arctan x}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}$, and find its domain of convergence.

Solution:

As $\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$ for every $-1 \leq x \leq 1$ then $\frac{\arctan(x)}{x} = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots$ for every $-1 \leq x \leq 1$ such that $x \neq 0$. As $f(0) = 1$ we conclude that $f(x) = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots$ for every $-1 \leq x \leq 1$. By the theorem of uniqueness of power series, we conclude that the Taylor series of f is $T_f(x) = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots$ and the range of convergence is $[-1, 1]$.

- iii. (5 points) Compute the value of $f^{(10)}(0)$.

Solution:

We have $\frac{f^{(10)}(0)}{10!} = -\frac{1}{11}$ (the coefficient of x^{10}). It follows that $f^{(10)}(0) = -\frac{10!}{11}$.

- (b) (8 points) Let $f : [0, 3] \rightarrow \mathbb{R}$ be a continuous function. Suppose that $\int_0^3 f(x) dx = 0$. Prove that there exists a point $0 < c < 3$ such that $f(c) = 0$.

Solution:

Consider the function defined by $F(x) = \int_0^x f(t) dt$ for every $x \in [0, 3]$. As f is continuous on $[0, x]$ then f is integrable on $[0, x]$ and therefore F is well-defined. Note that $F(0) = 0$ and that $F(3) = 0$. Also, by the fundamental theorem of calculus, F is differentiable and $F'(x) = f(x)$ for every $x \in [0, 3]$. Applying Rolle's theorem, then there exists a point $0 < c < 3$ such

that

$$f(c) = F'(c) = 0$$

3. (25 points)

- (a) (10 points) Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two sequences of real numbers. Suppose that $\sum_{n=1}^{\infty} a_n$ converges conditionally, and that $\sum_{n=1}^{\infty} b_n$ converges absolutely. Prove that $\sum_{n=1}^{\infty} (a_n - b_n)$ converges conditionally.

Solution:

First, as $\sum_{n=1}^{\infty} b_n$ converges absolutely then it converges, which implies that by the Algebra of series, $\sum_{n=1}^{\infty} (a_n - b_n)$ is convergent, as it is a difference of two convergent series. To prove that it converges conditionally, we must prove that it does not converge absolutely. ATC that $\sum_{n=1}^{\infty} |a_n - b_n| < \infty$. Then by the comparison principle for nonnegative series,

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} |a_n - b_n + b_n| \leq \sum_{n=1}^{\infty} (|a_n - b_n| + |b_n|) = \sum_{n=1}^{\infty} |a_n - b_n| + \sum_{n=1}^{\infty} |b_n| < \infty$$

in contradiction to the assumption that $\sum_{n=1}^{\infty} a_n$ converges conditionally.

- (b) (3 points) Prove that for every $p > 0$ and for every $2 \leq n \in \mathbb{N}$ we have

$$\frac{(-1)^n}{n^p + (-1)^n} = \frac{(-1)^n}{n^p} - \frac{1}{n^{2p} + (-1)^n n^p}$$

Note: You do not have to prove that the above expression is well-defined.

Solution:

Let $2 \leq n \in \mathbb{N}$ and let $p > 0$. We compute

$$\begin{aligned} \frac{(-1)^n}{n^p} - \frac{1}{n^{2p} + (-1)^n n^p} &= \frac{(-1)^n}{n^p} - \frac{1}{n^p (n^p + (-1)^n)} = \frac{(-1)^n (n^p + (-1)^n) - 1}{n^p (n^p + (-1)^n)} \\ &= \frac{(-1)^n n^p + 1 - 1}{n^p (n^p + (-1)^n)} = \frac{(-1)^n n^p}{n^p (n^p + (-1)^n)} = \frac{(-1)^n}{n^p + (-1)^n} \end{aligned}$$

- (c) (12 points) Let $\frac{1}{2} < p \leq 1$. Prove that the series $\sum_{n=2}^{\infty} \frac{(-1)^n}{n^p + (-1)^n}$ converges conditionally.

Solution:

First, note that the series $\sum_{n=2}^{\infty} \frac{(-1)^n}{n^p}$ is a Dirichlet series and therefore converges, and as $p \leq 1$ then it does not converge absolutely. Thus, it converges conditionally. The Series $\sum_{n=2}^{\infty} \frac{1}{n^{2p} + (-1)^n n^p}$ is a nonnegative series, since for every $p > 0$ and $n \geq 2$ we have $n^{2p} = (n^p)^2 > n^p$. Note that

$$L = \lim_{n \rightarrow \infty} \frac{\frac{1}{n^{2p} + (-1)^n n^p}}{\frac{1}{n^{2p}}} = \lim_{n \rightarrow \infty} \frac{n^{2p}}{n^{2p} \left(1 + \frac{(-1)^n}{n^p}\right)} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{(-1)^n}{n^p}} = 1$$

and as $2p > 1$ then the series $\sum_{n=2}^{\infty} \frac{1}{n^{2p}}$ converges (tail of $2p$ -harmonic), and by the limit comparison test, the series $\sum_{n=2}^{\infty} \frac{1}{n^{2p} + (-1)^n n^p}$ converges, and as it is nonnegative, it converges absolutely. By item (a), it follows that the series $\sum_{n=2}^{\infty} \frac{(-1)^n}{n^p + (-1)^n}$ converges conditionally.

4. (25 points)

- (a) (9 points) Let $a, b \in \mathbb{R}$ such that $a < b$. Let f be a function that is defined on $[a, b]$. Suppose that f is continuous on $[a, b]$

and let G be an anti-derivative of f in $[a, b]$. Prove that

$$\int_a^b f(x) dx = G(b) - G(a)$$

Solution:

See lectures.

(b) (16 points)

- i. (8 points) Prove that for every $a > 0$ there exists a $k \in \mathbb{N}$ such that for every $n \geq k$ we have $n! \geq a^n$.

Solution: Let $a > 0$. We have seen that the power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ has a convergence radius $R = \infty$. Thus, the series converges at the point $x = a$ and by the necessary criterion, we obtain that $\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$. It follows that there exists a $k \in \mathbb{N}$ for which $\frac{a^n}{n!} \leq 1$ for every $n \geq k$. Multiplying by $n!$ both sides of the inequality yields that $n! \geq a^n$ for every $n \geq k$.

- ii. (8 points) Let $(a_n)_{n=0}^{\infty}$ be a sequence of positive numbers. Let R be the convergence radius of the power series $\sum_{n=0}^{\infty} a_n x^n$ and suppose that $R < \infty$. Prove that the series $\sum_{n=0}^{\infty} a_n n! x^n$ diverges for every $x \neq 0$.

Solution:

Let $x > 0$. We choose a point $y \in \mathbb{R}$ such that $y > R$ (for example, $y = 2R$). By item (i), there exists a $k \in \mathbb{N}$ such that for every $n \geq k$ we have $n! \geq \left(\frac{y}{x}\right)^n = \frac{y^n}{x^n}$. Thus,

$$\sum_{n=k}^{\infty} a_n n! x^n \geq \sum_{n=k}^{\infty} a_n \frac{y^n}{x^n} x^n = \sum_{n=k}^{\infty} a_n y^n = \infty$$

By the comparison principle, we conclude that $\sum_{n=k}^{\infty} a_n n! x^n = \infty$ and therefore $\sum_{n=0}^{\infty} a_n n! x^n = \infty$ (as it has a diverging tail).

As the series diverges for every $x > 0$, it diverges on $(0, \infty)$ and we may conclude that $R = 0$, from which follows that the series diverges for every $x \neq 0$.

5. (25 points)

- (a) (15 points) Let $x_0, x \in \mathbb{R}$ such that $x_0 < x$ and let f be a function that is defined on $[x_0, x]$. Let $0 \leq N \in \mathbb{Z}$ and suppose that f is differentiable N times on $[x_0, x]$. Suppose also that $f^{(N)}$ is continuous on $[x_0, x]$ and differentiable on (x_0, x) . Let R_N be the Taylor remainder of f around x_0 of order N , and suppose that $R_N(x) = 0$.

- i. (8 points) Prove that there exists a point $x_0 < c < x$ such that $R_N^{(N+1)}(c) = 0$.

Solution:

Let T_N be the Taylor polynomial of order N around x_0 . As $R_N(t) = f(t) - T_N(t)$ for every $x_0 \leq t \leq x$ then R_N is N times differentiable on $[x_0, x]$ and $R_N^{(N)}$ is continuous on $[x_0, x]$ by AOC and differentiable on (x_0, x) by AOD. Also, $R_N^{(k)}(x_0) = f^{(k)}(x_0) - T_N^{(k)}(x_0) = 0$ for every $0 \leq k \leq N$. Thus, by Rolle 2, there exists a point $x_0 < c < x$ such that $R_N^{(N+1)}(c) = 0$.

- ii. (7 points) Conclude that $f^{(N+1)}(c) = 0$.

Solution: As T_N is a polynomial of degree $\leq N$ then $T_N^{(N+1)}(t) = 0$ for every $t \in \mathbb{R}$. Thus, for every $t \in (x_0, x)$ we have $R^{(N+1)}(t) = f^{(N+1)}(t) - T_N^{(N+1)}(t) = f^{(N+1)}(t)$. Plugging the value $t = c$ gives $f^{(N+1)}(c) = R^{(N+1)}(c) = 0$.

(b) (10 points) Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function that is defined by

$$F(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

- i. (3 points) Compute the value of $F'_x(0, 0)$.
- ii. (3 points) Compute the value of $F'_x(2, 2)$.
- iii. (4 points) Compute the value of $F''_{xy}(0, 0)$.

Solution:

(i) We have

$$F'_x(0, 0) = \lim_{x \rightarrow 0} \frac{F(x, 0) - F(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0$$

(ii) For every $(x, y) \neq (0, 0)$ we have

$$F'_x(x, y) = \frac{(x^2 + y^2)(y(x^2 - y^2) + 2x^2y) - 2x^2y(x^2 - y^2)}{(x^2 + y^2)^2}$$

Thus, $F'_x(2, 2) = 2$. (iii) We compute

$$F''_{xy}(0, 0) = \lim_{y \rightarrow 0} \frac{F'_x(0, y) - F'_x(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{-\frac{y^5}{y^4} - 0}{y} = \lim_{y \rightarrow 0} (-1) = -1$$